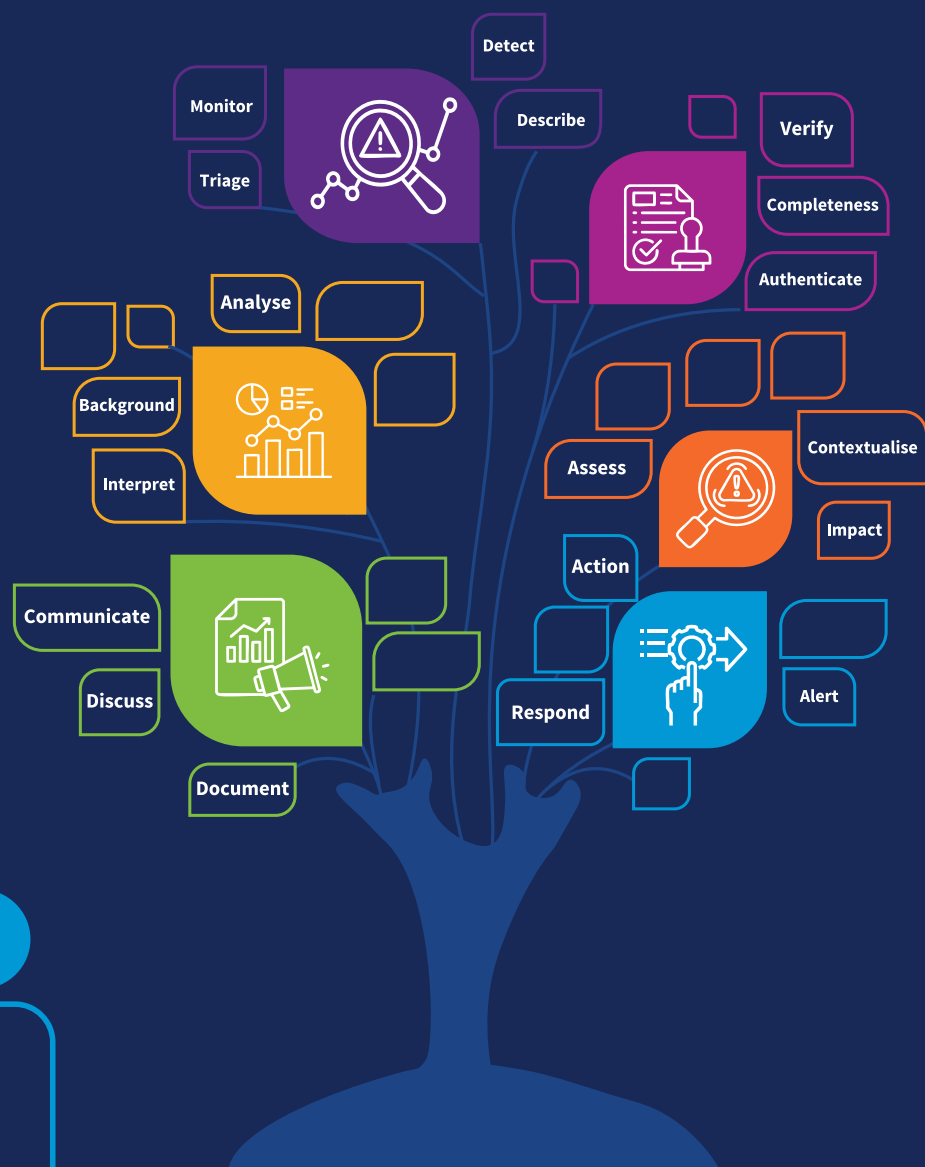


Public health intelligence curriculum



World Health Organization

HUB
Pandemic and Epidemic
Intelligence



Public health intelligence **curriculum**

Public health intelligence curriculum

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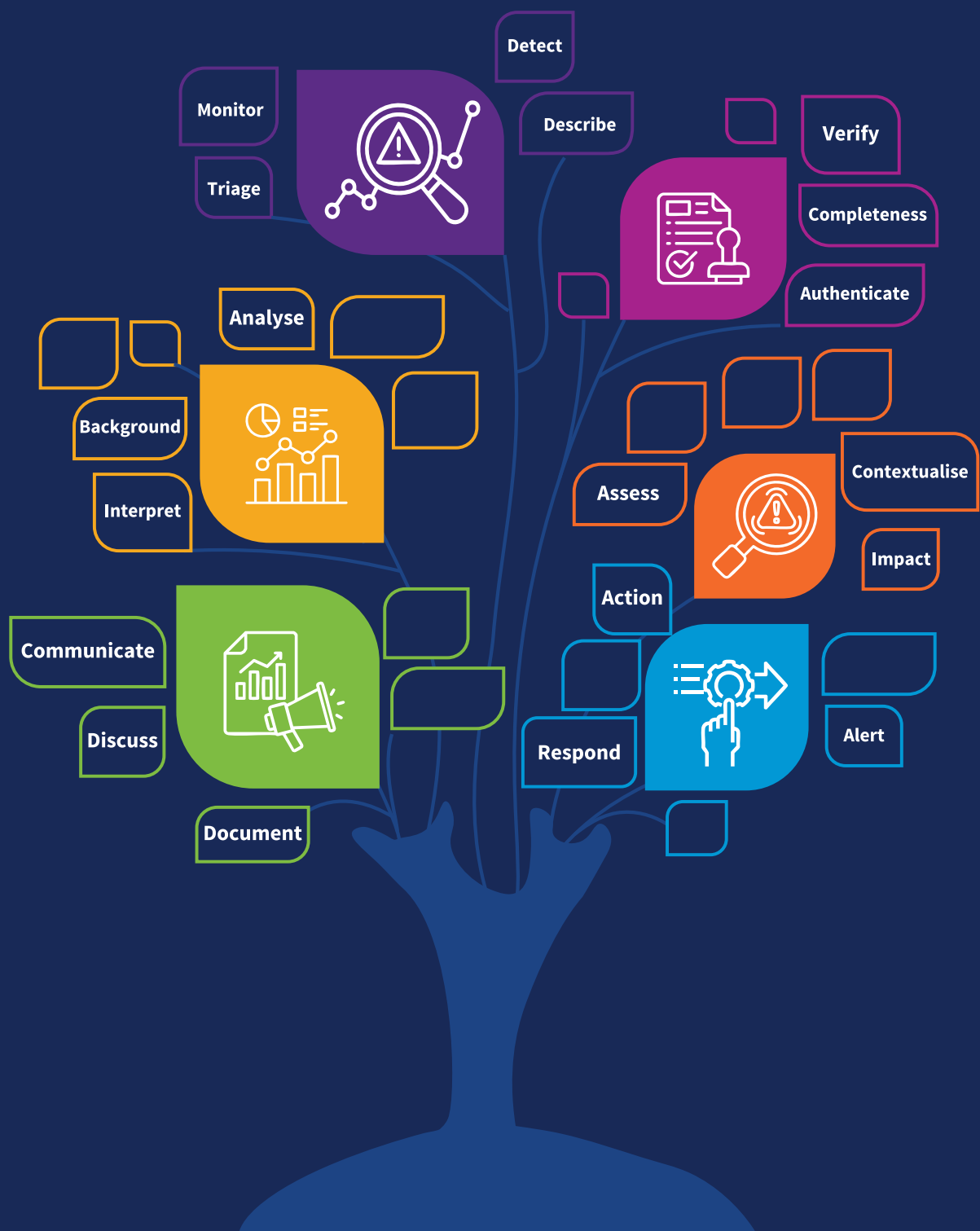
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Foreword

Public health challenges are becoming increasingly complex, shaped by globalisation, environmental pressures, technological change and the constant risk of emerging threats. Responding effectively requires a workforce that can translate information into timely, evidence-based action. Building these skills is central to strengthening public health intelligence and ensuring preparedness for the future.

The *Public health intelligence curriculum* provides a comprehensive, competency-based approach to developing these capabilities. Grounded in the *Public health intelligence competency framework*, it transforms defined competencies into practical learning opportunities through structured modules. These cover the full spectrum of public health intelligence – from signal detection and information verification, through analytical assessment, to the ethical and collaborative exchange of information.

Designed to be adaptable, the Curriculum can support training across diverse settings: national public health institutes, local health authorities, academic institutions and cross-sectoral initiatives. Its flexible structure allows for tailored learning pathways, ensuring relevance for those entering the field, advancing their expertise, or refreshing established skills. By embedding One Health principles, it also highlights the interdependence of human, animal and environmental health in confronting today's threats.

The development of this Curriculum has been a collaborative effort, drawing on the insights of experts, educators and practitioners around the world. We extend our sincere gratitude to all who contributed; colleagues and partners both within WHO and those across other organisations. Their commitment and insights have been central to shaping this document.

By applying this Curriculum, educators, trainers, policy-makers and institutions can help strengthen not only individual competencies, but also the collective capacity needed for a more coordinated, informed and resilient public health intelligence workforce. It is an investment in people and in a safer, healthier future.



A handwritten signature in black ink, appearing to read 'Chikwe Ihekweazu'.

Dr Chikwe Ihekweazu
Executive Director of the WHO
Health Emergencies Programme



Preface

Recent global events, from pandemics to climate-related emergencies and complex humanitarian crises, have exposed profound gaps and pressures in public health systems worldwide. These challenges have been intensified by rapid technological change, complex health threats and shifting societal expectations. Addressing them requires a public health intelligence workforce equipped not only with knowledge, but also with the practical competencies to act decisively and effectively. The *Public health intelligence curriculum* has been developed to meet this need, providing a structured pathway to build the skills essential for this critical field of practice.

Public health intelligence underpins sound decision-making in health policy and practice. It demands an integrated skillset: the ability to detect, verify, analyse and assess information, and to translate insights into timely action. In a time when data are abundant but not always reliable, the capacity to identify what matters, validate its accuracy, and use it to guide public health response has never been more vital. This Curriculum helps build those capabilities by transforming the *Public health intelligence competency framework* into practical, adaptable learning opportunities.

Organised around technical and functional domains, the Curriculum offers flexible modules that can be tailored to learners at different stages, from those entering the field to experienced professionals seeking to strengthen or expand their expertise. It embraces an all-hazards and One Health approach, recognising the interconnectedness of human, animal and environmental health, and the value of multisectoral perspectives in addressing a wide range of public health threats.

This Curriculum represents an investment in the continuous growth and excellence of the public health intelligence workforce. By fostering the competencies it sets out, we prepare professionals not only to meet current demands, but also to innovate, adapt and lead in protecting the health of populations. My hope is that it will serve as a practical, enduring yet evolving resource for educators and learners alike, strengthening global capacity and building a workforce ready for the challenges of today and tomorrow.



A handwritten signature in black ink, appearing to read 'Philip A. Malik'.

Dr Philip AbdelMalik,
Unit Head, Intelligence Innovation and Integration
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All external experts submitted to WHO a declaration of interest disclosing potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of this guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meeting or covered by the guidance.

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Executive Summary



Rapid identification of and response to health threats are critical for global health security. Recent major public health events have underscored the need to define essential public health intelligence (PHI) competencies to mitigate effects of such events and support countries to meet health regulation mandates. To address this need, the World Health Organization, in collaboration with global partners, has the Public health intelligence competency framework and Public health intelligence curriculum. The Framework aims to standardize the core elements and essential competencies of the PHI workforce.

The Curriculum builds upon the Framework by introducing a range of courses and learning objectives that collectively equip learners with the competencies, knowledge, skills and abilities that are essential for a skilled PHI workforce. The Framework and the Curriculum were developed through an iterative process involving research, stakeholder consultations and expert reviews.

The Curriculum serves as a training resource for organizations involved in public health surveillance, emergency responses, policy-making, education and workforce development. Its primary intended users are people involved in delivering or receiving PHI training and using preservice, in-training and in-service curricula. The Curriculum incorporates an all-hazards and One Health approach to ensure PHI professionals are equipped to handle a diverse range of health threats, including infectious diseases, environmental hazards and bioterrorism. The Curriculum is flexible and adaptable to various organizational and regional contexts, ensuring its relevance across different professional and geographical settings. It allows for customization based on audience expertise, organizational needs and resource constraints.

The Curriculum is organized into four technical domains – detect, verify, analyse and assess – with each technical domain covering core PHI competencies, and a functional domain focused on collaborative information exchange. These domains contain structured courses that address defined learning objectives for learners to develop the necessary competencies to prepare for real-world PHI responsibilities. In addition, the Curriculum presents an annex with a training module on the foundations of PHI that serves as an introduction to PHI for new members of the workforce or as a refresher for more established members.

The Curriculum is designed to evolve alongside advancements in data collection, analytics and public health methodologies. It is intended to be updated regularly to reflect advancements in PHI, emerging public health threats and evolving best practices. Through this, the Curriculum will provide a continually evolving foundational context for strengthening global PHI capacity, ensuring a well-trained workforce capable of addressing routine and emergent health threats.



Glossary

The following terms are defined within the context of public health intelligence and may differ in other contexts.

Actionable insights: findings from interpretation or preliminary assessment of data or information from multiple sources that lead to actions such as decision-making.

Alert: a signal that (i) has been verified to be an event, (ii) has been assessed and (iii) requires an intervention and collective action (an investigation, a response or a communication with partners or the public) (1). See also **Event, Signal** and **Verification**).

Alert threshold: a predefined criterion or set of criteria (such as the number of cases of a disease) that, when met or exceeded, triggers an alert, signifying a public health risk requiring response and action from authorities (1).

All hazards approach: an approach to the management of the entire spectrum of public health events based on the recognition that there are common elements and common capacities required in the management of these events, including in the responses to almost all emergencies (1).

Artificial intelligence (AI): a branch of computer science, statistics and engineering that uses algorithms or models to perform tasks and exhibits behaviours such as learning, making decisions and making predictions (2).

Attitude: a person's feelings, values and beliefs that influence their behaviour and the performance of tasks (3).

Behaviour: observable conduct towards other people or tasks that expresses a competency. Behaviours are measurable in the performance of tasks (3).

Capability: the potential and capacity of an individual to perform specific tasks. Capability involves adaptability, judgment and future-focused thinking (3). See also **Competency**.

Collaborative surveillance: the systematic strengthening of capacity and collaboration among diverse stakeholders, both within and beyond the health sector, with the goal of enhancing public health intelligence and improving evidence for decision-making. Collaborative surveillance provides the enabling environment for public health intelligence (4).

Competency: the ability of a person to integrate knowledge, skills and abilities in their performance of tasks in each context. Competencies are durable, trainable and measurable (3). Competencies are the measures of how an individual performs a capability. See also **Capability**

Competency framework: an organized and structured representation of a set of interrelated and purposeful competencies that are needed to perform a job well (3).

Detection: a process by which signals of potential public health concern are identified, recorded and preliminarily assessed (5).

Early warning, alert and response (EWAR): an organized process to rapidly detect and respond to signals during acute public health events of any origin (1).

Epidemic Intelligence: Early warning, alert and response (EWAR): an organized process to rapidly detect and respond to signals during acute public health events of any origin (1).

Event: defined by the International Health Regulations as “a manifestation of disease or an occurrence that creates a potential for disease” (7). It may include events that are infectious, zoonotic, food safety, chemical, radiological or nuclear in origin; and that may be transmitted by persons, vectors, animals, goods or food, or through the environment. In the context of this framework (and early warning, alert and response), an event refers to a signal that has been verified (1). See also **Alert, Signal** and **Verification**

Event-based surveillance: the organised collection, monitoring, assessment and interpretation of mainly unstructured ad hoc information regarding events or hazards that may represent an acute threat to human health (8). Event-based surveillance supports public health intelligence by feeding signals from diverse sources into the intelligence process.

Global public health security: the activities required, both proactive and reactive, to minimize the danger and impact of acute public health events that endanger people's health across geographical regions and international boundaries (9).

Hazard: a process, phenomenon, or human activity that may cause loss of life, injury, or other health impacts; property damage; social and economic disruption; or environmental degradation (10). Hazards may be single, sequential, or combined in their origin and effects. Under WHO classification, hazards include natural hazards (biological, extraterrestrial, geophysical and hydro-meteorological); human induced hazards (technological and societal); and environmental hazards (environmental degradation). Hazard focuses on the source of the harm rather than its likelihood. Further descriptions and classifications of hazards have been defined by the United Nations Office for Disaster Risk Reduction (11). See also **All-hazards approach**.

Indicator-based surveillance: a routine system for collecting, analysing and interpreting data on reportable diseases. The data are structured according to case definitions (12).

Integrated surveillance: the use of unified methods to conduct surveillance for multiple public health risks (13).

International Health Regulations: a legal instrument of international law that is legally binding on 196 countries, including the 194 WHO Member States. The Regulations aim to help the international community prevent and respond to acute public health threats that have the potential to cross borders and threaten people worldwide (7).

Knowledge: the recall of specifics and universals; the recall of methods and processes; and/or the recall of a pattern, structure or setting (3).

Learning objectives: the aims that should be achieved by the end of a learning experience. Learning objectives reflect change towards a specified standard or level. Course learning objectives capture the knowledge, skills and abilities related to the subject matter of the course (3,14).

Machine learning (ML): the subset of artificial intelligence that allows computer algorithms to learn through data, without being explicitly programmed to perform a task (2).

One Health approach: an integrated, unifying approach that aims to sustainably balance and optimize the health of people, animals and ecosystems. The approach recognizes that the health of humans, domestic and wild animals, plants and the wider environment (including ecosystems) are closely linked and interdependent. It mobilizes multiple sectors, disciplines and communities at various levels of society to work together to foster well-being and tackle threats to health and ecosystems. It also addresses the collective need for clean water, energy and air; safe and nutritious food; action on climate change; and contributions to sustainable development (15).

Public health intelligence: a core public health function responsible for identifying, collecting, connecting, synthesizing, analysing, assessing, interpreting and generating a wide range of information for actionable insights; and disseminating them for informed and effective decision-making to protect and improve the health of the population (16).

Public health intelligence curriculum: a document that delineates the philosophy, goals, objectives, learning experiences, instructional resources and assessments that comprise a specific educational programme. The Curriculum encompasses the organization and sequencing of content necessary to develop the knowledge, skills and abilities needed to carry out core public health intelligence functions, and advice on modifying content to target audiences and selecting learning activities and assessments to ensure training effectiveness.

Public health intelligence Data: data collected from a variety of sources encompassing information from human health, human non-health and non-human sectors on existing or emerging public health threats (5).

Risk: the likelihood (or probability) of a specific event occurring within a given time if a person is exposed to a threat or hazard. Risk is a quantifiable concept that often reflects the likelihood of a negative event, such as death or disease (17).

Risk assessment: a continuous process of gathering, assessing and documenting information to estimate the nature and probability to adverse effects in humans, animals or the environment following an event or exposure to a hazard. A risk assessment provides the basis to inform the actions to reduce the negative consequences of public health events through preparedness and response activities (18).

Risk assessment tool: a standardized tool used to visually evaluate and prioritize potential health hazards by categorizing them based on likelihood of occurrence and severity of impact, allowing public health officials to identify which risks require the most immediate attention and mitigation strategies (19).

Risk communication: the real-time exchange of information, advice and opinions among experts, community leaders or officials and the people who are at risk or who have a direct influence on risk mitigation due to their practices or behaviour. Risk communication ensures people and communities are aware of current threats and can be used to promote behaviours to reduce risks (18,19).

Signal: the initial detection of a potential public health event, prior to verification. Signals may consist of information or reports of cases or deaths (individual or aggregated); potential exposure of humans to biological, chemical or radio nuclear agents; or the occurrence of natural or human-induced disasters (8). Signals can be raw, not yet assessed, in verification or verified. A verified signal transitions into an event. See also **Alert, Event, and Verification**

Skill: a specific cognitive or motor ability that is typically developed through training and practice and is not context-specific (3).

Social listening: monitoring the understanding, questions, concerns, information voids, narratives, misinformation, and disinformation that circulate in web-based and offline environments (20).

Surveillance (public health surveillance): the systematic, ongoing collection, collation and analysis of data for public health purposes and the timely dissemination of public health information for assessment and public health response, as necessary (7,10).

Task: an observable unit of work that draws on knowledge, skills and abilities. Tasks are time-limited, trainable and measurable (3).

Threat: a situation or development, or a combination of these, that can harm health security, as either a real or perceived danger (8). Examples include antimicrobial resistance or environmental threats such as climate change. Threats can also refer to deliberate events such as an intent to release a hazardous substance to cause harm (21).

Triage (of data and signals): the process of identifying, sorting and prioritizing data and addressing their quality issues (1).

Triangulation (of data): the process of combining and analysing public health data from multiple sources to generate a more comprehensive and accurate understanding of public health signals. This technique helps to cross-verify findings, reduce bias and identify patterns or trends that might not be evident when relying on a single data source (1,22).

Verification: confirming the authenticity, completeness and accuracy of the data, and eliminating hoaxes, false rumours and artefacts. Verification involves double-checking the primary source, integration and comparison with data from other sources, comparing with specified parameters and indicators, or performing field investigations (1). See also **Triangulation (of data)**.



Introduction

Background

The ability to rapidly identify and respond to health threats is critical to global health security. At the World Health Assembly in 1969, the International Health Regulations were adopted, providing a legal framework that requires World Health Organization (WHO) Member States to strengthen and maintain core capacities to detect, assess, notify and report public health events within their territory. The International Health Regulations have since undergone amendments and, more recently, a substantial revision, which was adopted in 2005 and entered into force in 2007 (7).

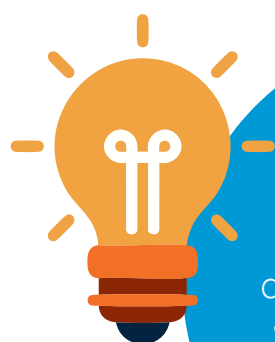
Public health intelligence (PHI) is a key component of achieving the International Health Regulations. PHI is defined as “a core public health function responsible for identifying, collecting, connecting, synthesizing, analysing, assessing, interpreting and generating a wide range of information for actionable insights; and disseminating them for informed and effective decision-making to protect and improve the health of the population” (5,16). PHI depends on timely data, comprehensive systems to collect the data, and people with the necessary skills and judgement to identify potential threats. Developing and sustaining PHI requires a team with the professional competencies to transform an abundance of raw data, from a variety of sources, into insights, information and knowledge to drive effective and confident public health decisions and actions.

It has become apparent that there is a strong need to develop an overarching competency-based training structure for the PHI workforce. Ultimately, this will allow enhancement of global health emergency prevention, preparedness, response and resilience by strengthening PHI capacity, ensuring a global pool of well-trained people for PHI work in the health threat landscape.

In 2023, WHO launched the Collaborative Surveillance Framework, which was developed as a core component of the WHO Health Emergencies Programme (4). This was followed by the development of the Competencies for One Health Field Epidemiology (COHFE) Framework, as a collaboration between WHO and the Food and Agriculture Organization (23). These two frameworks aim to improve PHI and decision-making and enhancing core One Health and sector-specific knowledge, skills and competencies for field epidemiologists.

Public health intelligence (PHI) is defined as “a core public health function responsible for identifying, collecting, connecting, synthesizing, analysing, assessing, interpreting, and generating a wide range of information for actionable insights; and disseminating them for informed and effective decision-making to protect and improve the health of the population” (5,16).

Building on the concept of multisectoral collaborations, WHO have recently developed the Public health intelligence competency framework which is accompanied by this Public health intelligence curriculum. Together, these products outline core PHI components and the essential competencies and training needed by the PHI workforce to effectively perform them, using an all-hazards and multisectoral approach. In the absence of internationally accepted PHI competencies, the PHI Framework and Curriculum aim to illustrate the ideal functions and tasks for effective PHI to allow for better collaboration and coordination across organizations.



The Curriculum
introduces different
courses that **support**
the training of the
PHI workforce

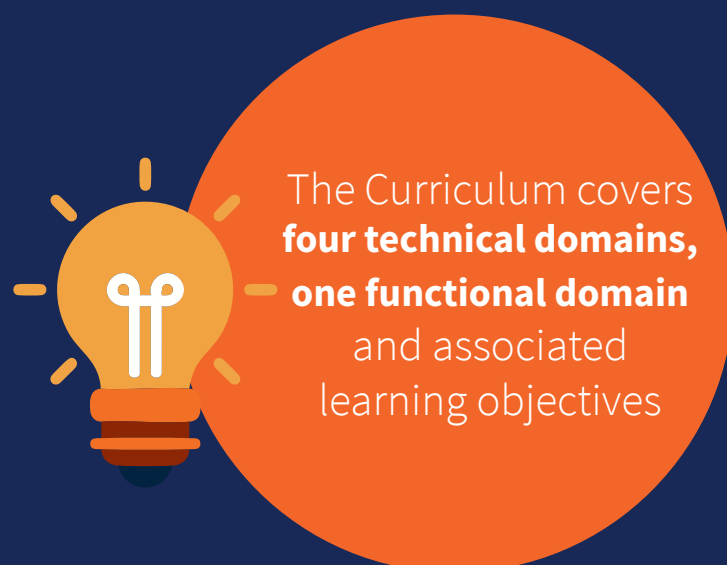


Development of the Curriculum

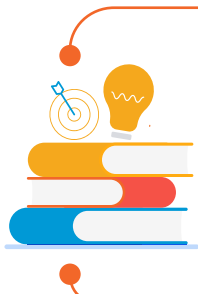
The Curriculum was created as a companion to the Public health intelligence competency framework. Both documents went through an iterative development process of research, drafting and review. This included a landscape analysis of existing PHI training materials; stakeholder consultation; review of job listings; and review of existing relevant frameworks, curricula and literature. The Curriculum and Framework are intended to complement related resources such as the Competencies for One Health Field Epidemiology (COHFE) Framework (24) and the Global Competency Framework for Universal Health Coverage (3).

Following its development, the Curriculum underwent several rounds of review and approval from subject matter experts, including the PHI Training Working Group (composed of technical experts from organizations around the world, including the Africa Centres for Disease Control and Prevention, the Centers for Disease Control and Prevention, the European Centre for Disease Prevention and Control, the Food and Agriculture Organization, the Robert Koch Institute and WHO) and other experts from several organizations.

To ensure the Curriculum would provide the competencies, knowledge, skills and abilities needed to perform the core functions of PHI, its contents were aligned with the Framework through a set of courses. These courses were associated with specific learning objectives following the revised Bloom's taxonomy (25). The additional training course on foundations in PHI provides foundational knowledge and skills applied across all the courses in the Curriculum.



Scope



The Curriculum is designed to apply to PHI work conducted in a variety of organizations with various scopes of work, including different geographical scopes and extents of public health threats (e.g. human or animal infectious diseases, chemical or nuclear hazards).

In the context of the Curriculum, PHI covers intelligence activities that support health protection and global public health security (25). The goal is for the Curriculum to support training of PHI personnel across a range of activities, including data detection and characterization, verification, analysis and assessment, and regular communication with internal and external collaborators and stakeholders. The Curriculum can be used in the creation of training materials in new programmes, and in the review and revision of training materials used by established programmes.

The Curriculum takes an all-hazards and multisectoral (including One Health) approach to PHI. The all-hazards approach acknowledges that although hazards vary in source, they challenge health systems in similar ways, demanding a multisectoral response. This aligns with One Health, which recognizes that improved coordination between the human, animal and environmental health sectors has reciprocal benefits and leads to unifying, integrated and sustainable solutions to health problems (15,26). Using an all-hazards and One Health approach recognizes how PHI benefits from multisectoral collaboration and understanding, because of the many different types of threat facing public health. These threats include infectious diseases, natural or human-initiated disasters, and intentionally or unintentionally released chemical, biological, radiological or nuclear agents or materials (21). It is important for countries to rapidly identify all such health threats to be able to quickly initiate response measures.

Although the Curriculum covers training for the workforce involved in PHI across a broad range of roles, it does not cover every role or training situation that the workforce involved in PHI might encounter. Organizations are encouraged to add or remove learning objectives as required to meet the specific needs of the target audience and the materials being developed.

This document is a first-release version. It is intended to be revised and updated as needs and experiences change and other PHI official products are developed. This version of the Curriculum reflects the current stage of PHI knowledge, requirements and processes, with the acknowledgement that these are likely to change with rapid advances in data collection methods and analytical techniques and the growing availability of real-time health data from a wide range of sources.

Structural content

The structural content of the Curriculum is based on the process that takes place within an established PHI system (18, 24). The process consists of several steps in which signals are detected, verified, analysed and assessed for risk (Figure 1). This is followed by a separate step of further action and response, which is not covered in the Curriculum.

In the Curriculum, the first four steps of the PHI process are presented through technical domains that describe specific technical tasks undertaken in each step:

Technical domain 1: detect	It involves routine monitoring of data and signals and their key related information. The data and signals undergo triage to determine the action that needs to be taken, such as discarding, further monitoring or proceeding to verification.
Technical domain 2: verify	It involves confirming the authenticity and accuracy of the signals and the credibility and reliability of the information source. This is a collaborative process that involves communication and discussion with different stakeholders and consultation with official sources.
Technical domain 3: analyse	It involves epidemiological and statistical analyses and their interpretation. Consultation outside the PHI team and organization is often required. Analysis activities may also take place during other steps of the PHI process as new information becomes available.
Technical domain 4: assess	It involves determining the significance of the signals or event (for verified signals) and assessment of risk to the exposed population, multisectoral impact, the efficacy of response measures and overall generating actionable insights.

These four technical domains have been selected because they represent the steps in which possible public health threats are identified early and clearly assessed and understood, to provide solid evidence for decision-making and further action. During the performance of these steps, the PHI workforce should be able to clearly demonstrate the PHI competencies.

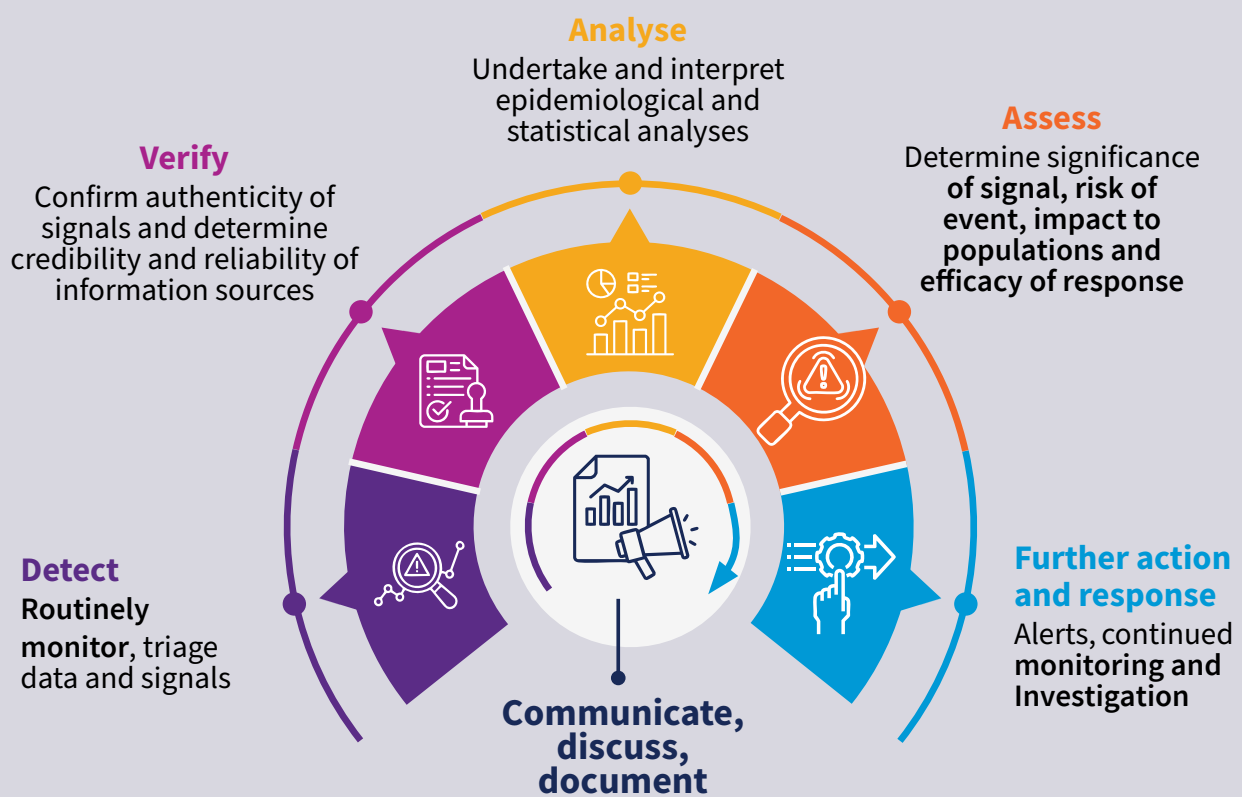
The technical domains also encompass an overarching step that focuses on regular communication, discussion and documentation as new information becomes available throughout the process. This occurs continuously at any time during the PHI process and includes interaction between internal team members, external collaborators and all other relevant stakeholders.

In general, the PHI process may be perceived as having a cyclical nature. The steps in the PHI process are not necessarily sequential, however, and may be skipped or repeated, depending on the circumstances. For example, signals of an emerging infectious disease may require immediate further action and alerts during the initial steps of analysis and assessment. Steps may also be combined, such as when the “detect”, “verify” and “analyse” steps are undertaken by the same team in an organization and the “assess” step is undertaken by another team. The technical domains in the Curriculum, representing

the steps in the PHI process, are presented in a chronological order to facilitate simpler, more structured understanding.

It is worth noting that different organizations and different guidelines may use different terminology for the steps in the PHI process. The overall goal of the PHI process is to move from data through knowledge synthesis to action, with the specific aim of early detection for timely response. It is important to recognize that the PHI process presented in the Curriculum is generalized and can be modified to individual situations or institutions and their specific requirements.

Figure 1. A simplified representation of the public health intelligence (PHI) process

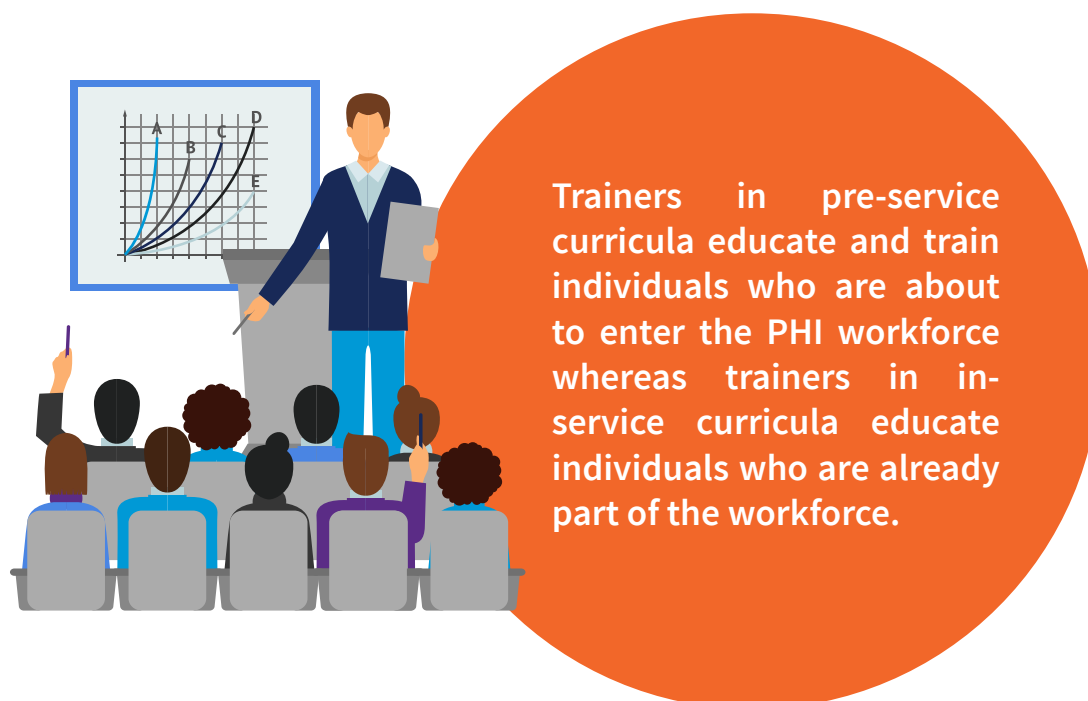


Source: Adapted from the WHO Epidemic Intelligence from Open Sources (EIOS) initiative (27)

How to use the Public Health Intelligence Curriculum

Training and Learning audience

The Curriculum is aimed at two target user groups: trainers and learners. Trainers are those involved in delivering training to the PHI workforce, including preservice curricula (e.g. in higher education), in-training curricula (e.g. field epidemiology training programmes) and in-service curricula (e.g. in applied public health settings).



Trainers in pre-service curricula educate and train individuals who are about to enter the PHI workforce whereas trainers in in-service curricula educate individuals who are already part of the workforce.

Preservice curricula are aimed at people who are about to enter the PHI workforce, and in-service curricula are aimed at people who are already part of the workforce. In addition, the Curriculum may be relevant for policy-makers, licensing and regulatory agencies, and governmental, nongovernmental and private-sector organizations.

Learners are the workforce involved in PHI operating at all levels. These people may have explicitly labelled PHI roles (e.g. PHI officers), but others may not (e.g. human or animal health epidemiologists and surveillance officers, environmental health officers, food safety professionals). The Curriculum can support training both in organizations with limited funding or capacity for PHI and in organizations where PHI programmes are well established, funded and resourced. Implementation of the Curriculum can take place in a range of settings, including in-person, virtual, hybrid and self-study, depending on the context and specific needs for achieving the learning objectives.

Modular approach to course adaptation

The Curriculum was developed for a broad target audience – for example, it can be used to create a preservice university course for masters students or an in-service training course for experienced professionals. Organizations using the Curriculum are likely to vary in their mission and have different audiences, and it may be necessary to adapt the Curriculum to meet distinct needs. The Curriculum has been developed as a modular manual that allows for various levels of adaptation.

In the context of adaptation, it should be noted that training in foundations of PHI serves as a base for the courses presented in the four technical and functional domains (see Annex 1). This covers introductory knowledge and core skills used throughout the Curriculum. When the training in foundations of PHI has been completed (whether through the Curriculum or other training), participants can use a building-block approach to complete selected courses in the Curriculum, driven by the needs of the organization and combined with the desired competencies, knowledge, skills and abilities to be developed. The Curriculum can be adapted through a step-by-step process or take a less ordered, more fluid approach.

Figure 2 outlines the general steps that should be followed to create a training programme. The target audience and goals of the training should be determined, followed by several steps in which individual roles and wider context are analysed, with the overall purpose of informing which technical and functional domains and courses should be included. At this point, the desired training is developed.

Figure 2: Stepwise guide for adapting the Public health intelligence curriculum to customized training



01

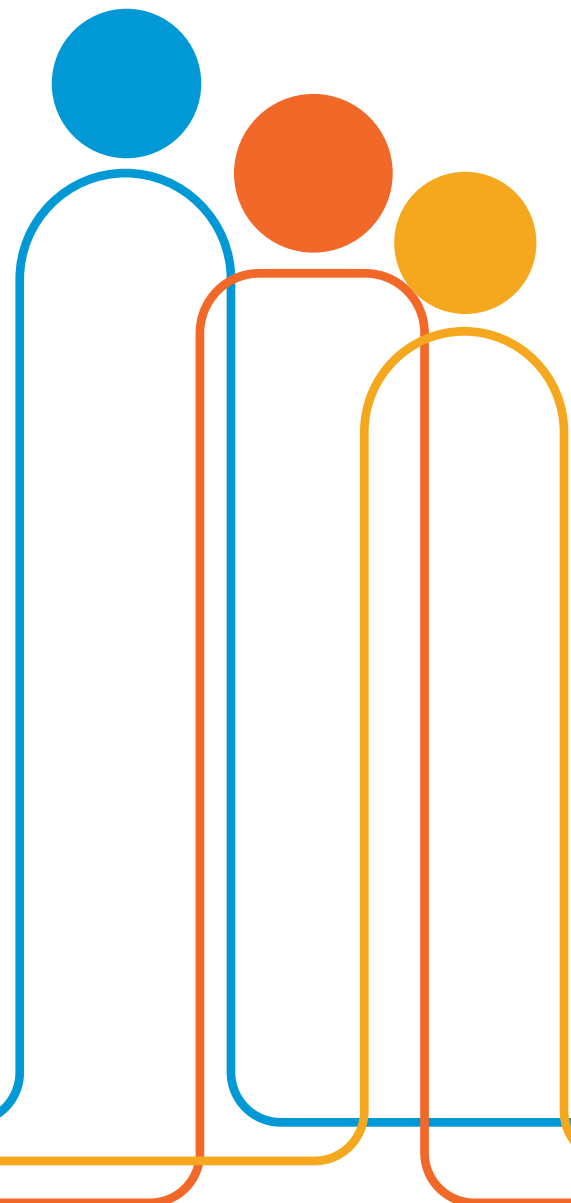
When adapting the Curriculum, first consider the scope of the training. If a comprehensive PHI training is to be developed, it is recommended that all parts of the Curriculum are used. If the training is intended for a specific subset of the workforce involved in PHI, a subset of the Curriculum can be used (i.e. the courses and learning objectives that are applicable to people filling that role). PHI activities vary between organizations, and workforce members may hold PHI roles that cover several functions – to develop training for these people, it is recommended to review the Framework tasks. This allows determination of the tasks that the people will conduct and, subsequently, identification of the courses and learning objectives in the Curriculum that best align with those tasks.

02

Next, consider the level of experience, knowledge and expertise of the learning audience. People with extensive PHI experience may need only refresher training in certain areas. Building training for this audience could involve selecting a specific technical domain or course and removing topics that are already performed well and lower-level learning objectives. To create training for an audience with limited knowledge and experience, all technical domains and courses relevant to their roles may be necessary.

03

The final consideration is the context of the training, accounting for aspects such as language, culture and available resources. The materials in the Curriculum should be adapted to the context. Such adaptations may include translations, modifying wording or selecting specific content to respect cultural differences. When the budget or time is constricted, some technical domains and courses may need to be prioritized to ensure the most critical concepts are conveyed.



Technical and functional domains, courses, and associated competencies and learning objectives

The Curriculum comprises four technical domains – detect, verify, analyse, assess – and one functional domain – collaborative information exchange (Figure 3). Each domain is divided into courses and associated learning objectives that instil competencies (Table 1), knowledge, skills and abilities. In addition, each technical domain has an overarching course that covers details of training related to communication, discussion and documentation.

In addition to the technical and functional domains, the Curriculum introduces training on the foundations of PHI (Annex 1). This training aims to prepare learners for roles in PHI by exploring the historical foundations, practical applications, ethical considerations and legal frameworks that underpin PHI activities. Generally, it is recommended that all learners, irrespective of their PHI roles and level of experience, complete the training in the foundations of PHI because it can serve as a first presentation or as a refresher of PHI concepts.

Figure 3 includes the Public health intelligence competency framework, which was used as the foundation for creating the Curriculum. The Framework emphasizes competencies and tasks that are essential for effective performance by the PHI workforce. The Framework has a similar structure to the Curriculum, centred around technical and functional domains. Through completion of the courses and associated learning objectives in the Curriculum, learners gain the knowledge, skills and abilities needed to perform the tasks presented in the Framework and strengthen their corresponding competencies (Table 1).

Figure 3. Content of the PHI Curriculum and relation to the Competency Framework

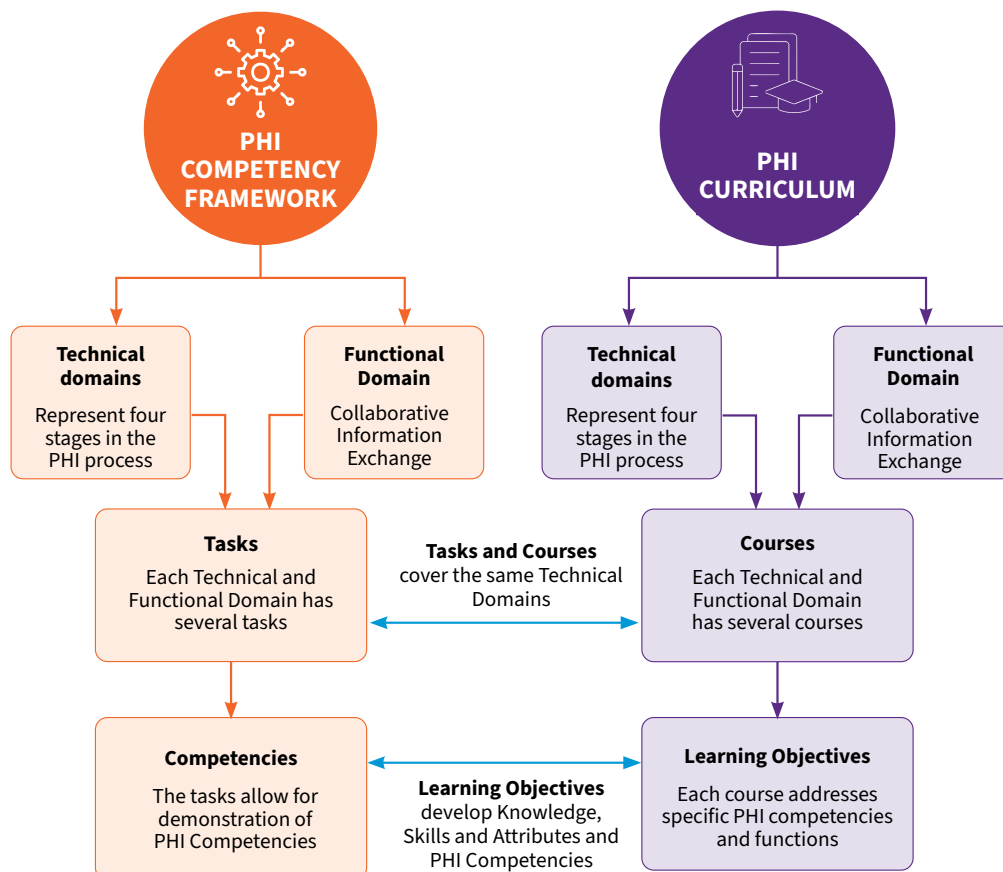


Table 1. Technical and functional competencies in the public health intelligence (PHI) process^a

Technical Competencies
Competency 1: Assesses data and information from a range of sources
Competency 2: Engages in practices to ensure data security, confidentiality and privacy
Competency 3: Considers contextual factors in the analysis of PHI data
Competency 4: Contributes to a culture of continuous quality improvement of PHI data
Competency 5: Takes and adaptive, collaborative and rigorous approach to PHI decision-making
Competency 6: Incorporates a systems approach to decision-making, considering One Health and all-hazards perspectives
Competency 7: Takes a solutions-oriented approach to problem solving in PHI
Competency 8: Demonstrates effective use of data for generating actionable insights
Competency 9: Adapts priorities based on unexpected or changing situations
Competency 10: Engages multisectoral One Health stakeholders for implementing joint PHI activities
Competency 11: Builds and maintains a trusting network of multisectoral One Health stakeholders to support PHI activities
Competency 12: Learns from a range of multisectoral One Health partners
Competency 13: Incorporates a holistic approach to PHI
Functional Competencies
Competency 14: Ensures others have clear, accurate, and timely information
Competency 15: Proactively manages interactions with others
Competency 16: Adapts communication to the goals, needs, cultural context, urgency, and sensitivity of the interaction
Competency 17: Listens actively and attentively
Competency 18: Conveys information purposefully
Competency 19: Acknowledges the importance of stakeholder information needs
Competency 20: Tailors communication to the audience
Competency 21: Works within the limits of competence and scope of practice for PHI activities
Competency 22: Engages in lifelong learning and reflective practice to improve performance of PHI activities
Competency 23: Demonstrates high standards of ethical conduct
Competency 24: Focuses on well-being, recognises stress and seeks support when needed
Competency 25: Operates with cultural sensitivity, respectfulness, and compassion
Competency 26: Adapts to unexpected or changing situations
Competency 27: Plans strategically
Competency 28: Empowers others
Competency 29: Manages tensions and conflicts

^a Each competency has a list of associated behaviours used to demonstrate competencies (see Annexes 1 and 2). In relation to the Public health intelligence competency framework, behaviours refer to the practical way in which a PHI workforce member applies competencies to complete the tasks and subtasks in the technical or functional domains.

Technical Domains



Technical Domain 1: Detect



Introduction



This step is the process of identifying and understanding the signals that can indicate potential public health events, setting the foundation for a timely, effective response. Through the courses in this domain, learners explore detecting signals and documenting key information related to identification of the signals. Learners gain insight into triage of signals, using triangulation approaches to determine whether they are real signals. Learners are introduced to the process that occurs when deciding whether signals should be discarded, monitored further (shared for awareness) or proceeded to verification.

Note that although signals may be discarded in this step, signals may also be discarded at later stages as more information is identified, analysed and verified.

Competencies

The most relevant core PHI technical competencies for this technical domain are:

Competency 1:
Evaluates data and information from a range of sources



Competency 2:
Engages in practices to ensure **data security, confidentiality and privacy**



Competency 4:
Contributes to a culture of **continuous quality improvement** of PHI data



Courses

There are three separate courses and one overarching course in this technical domain. Each separate course has several learning objectives (Table 2), which collectively target the competencies that learners gain in this technical domain:

Overarching course T1.0 – communicate, discuss and document

Learners discover how to prepare for and organize the communication and documentation process, including identifying stakeholders, identifying and preparing relevant protocols, and setting up templates used for documentation. Learners are introduced to methods for discussing the results of the triage, including when to discard, monitor further or proceed to verification of signals.

Course T1.1 – detect signals from PHI data

This provides an overview of monitoring, which is an essential step in PHI. Learners discover how to identify various sources of information, within and beyond the human health sector, and describe the type of information each source can provide. The topic delves into strategies for incorporating indicator-based and event-based surveillance into systematic monitoring processes to identify emerging trends and signals.

Course T1.2 – describe signals from PHI data

This provides a foundational understanding of descriptive epidemiology to describe signals. Learners explore how to apply descriptive epidemiology concepts to interpret information from various sources. Learners develop skills in consolidating information from multiple sources to construct comprehensive threat descriptions. Emphasis is placed on identifying potential sources of additional information and integrating supporting data to enhance understanding of the signal. Additionally, learners discover how to differentiate signal types.

Course T1.3 – triage signals from PHI data

This equips learners with essential skills for efficiently triaging public health signals. Learners discover how to distinguish genuine changes from artefacts, interpret data trajectories and determine the urgency of identified signals. Learners explore the application of the International Health Regulations (2005) and how to determine the outcome of triage (discard, monitor further or verify).

Table 2: Courses and learning objectives in technical domain 1 - detect

Courses	
Communicate, discuss and document	
Learning objectives	Describe how to identify stakeholders who need to be informed
	List sources of relevant protocols and describe methods for creating templates to document all steps
	Explain how to discuss triage outcomes and decide on next steps
Detect signals from PHI data	
Learning objectives	Identify official and unofficial quantitative and qualitative data sources directly related to human health
	Identify official and unofficial quantitative and qualitative data sources indirectly related to human health
	Demonstrate how to conduct media monitoring, including mainstream media, social media and community sources for social listening
	Describe the type of information each source can provide
	Assess the quality of information sources
	Explain how to incorporate indicator-based and event-based surveillance into regular PHI data monitoring
	Use software and systems for managing data, signals and additional information
	Distinguish between facts, context-specific evidence, credible information and misinformation
Describe signals from PHI data	
Learning objectives	Describe the clinical, epidemiological and laboratory features of known health threats
	Describe potential deliberate (unnatural) means of delivery of health threats
	Determine the epidemiological characteristics of the signal using descriptive epidemiology
	Describe how to identify information from health and non-health sources that describes known and unknown facts about the signal
	Identify geopolitical events that could impact the signal
	Identify sociological and cultural factors that could impact the signal
	Integrate supporting information that may be useful to create a more complete description of the signal
	Identify what is known, uncertain or unknown about the threat based on the information available
Triage signals from PHI data	
Learning objectives	Explain methods for triaging signals
	Explain where to find baselines, thresholds and seasonal trends for common hazards
	List potential sources of additional information that can support data triangulation
	Describe how to perform data triangulation by consolidating data and information from multiple sources
	Distinguish real changes from artefacts in a signal
	Apply Annex 2 of the International Health Regulations (2005) to determine whether a signal may constitute a public health emergency of international concern
	Describe common reasons for discarding signals during the triage process
	Determine whether a signal should be monitored, verified or discarded

Technical Domain 2: Verify



Introduction



Verifying PHI signals is a critical process that aims to confirm their authenticity and accuracy. It includes ensuring the source of information is accurate, credible and reliable. Through the courses and learning objectives in this technical domain, learners become familiar with methods for signal verification, including engaging stakeholders and using trusted official sources. Learners gain skills in effective communication to gather and assess information, ultimately determining the verification status of signals.

A verified signal constitutes a public health event. It is important to note, however, that the event may fall outside a given organization's scope or mandate.

Competencies

The most relevant core PHI technical competencies for this domain are:

Competency 1:
Evaluates data and information from a range of sources



Competency 4:
Contributes to a culture of **continuous quality improvement** of PHI data



Competency 8:
Demonstrates **effective use of data** for generating actionable insights



Courses

There is one separate course and one overarching course in this technical domain. The separate course has several learning objectives (Table 3), which collectively target the competencies that learners gain in this technical domain:

Overarching course T2.0 – communicate, discuss and document

This focuses on using official communication channels to maintain regular contact with all stakeholders and to engage with other collaborators that can confirm the accuracy of the signal. Learners explore how to communicate about the verification process and how to discuss the validity of the signals. Learners are introduced to methods for documenting the verification process and to the importance of storing data in a secure location.

Course T2.1 – verify the completeness, consistency and significance of PHI signals:

This course equips learners with essential skills for determining whether the signal is accurate, complete and consistent with other information sources. Learners explore how to consider further information in the local and wider contexts, through official channels or sources. Learners gain skills in effective communication to gather and assess information, including from the original source of the signal, ultimately determining the verification status of signals.

Table 3: Courses and learning objectives in technical domain - verify

Courses	
Communicate, discuss and document	
Learning objectives	Describe how to identify established communication channels
	Describe how to engage stakeholders who may have information that can be used to verify a signal
	Explain how to identify colleagues, experts and stakeholders who can share information about the original source and additional sources of information
	Identify the best approaches for discussing the significance of the signals
Verify the completeness, consistency and significance of PHI signals	
Learning objectives	Explain methods for confirming the accuracy and credibility of signals, including adherence to the International Health Regulations (2005)
	Explain epidemiological approaches to signal verification
	Identify trusted official sources of information that can be used to verify a signal
	Describe how to obtain additional information from stakeholders to verify a signal
	Explain how to determine whether the signal is verified based on communication with stakeholders
	Describe the next steps following signal verification and classification of the signal as an event

Technical Domain 3: Analyse



Introduction



This topic provides an advanced exploration of the analysis and interpretation of signals from PHI data. Learners discover how to identify and use various sources of information for comprehensive signal analysis, including statistics and spatial analysis tools. Learners apply epidemiological and statistical principles to interpret trends, identify at-risk populations and predict changes, while recognizing the limitations of epidemiological analyses. Learners explore interdisciplinary collaboration and contextual factors influencing PHI data analysis.

It is important to note that activities in this technical domain may also take place throughout the “detect” and “verify” steps in the PHI process as data and information are gathered and updated. This varies between organizations and settings, however, and “analyse” is presented as a separate step in the Curriculum, while acknowledging that its activities in many circumstances happen concurrently with the initial steps in the PHI process.

Competencies

The most relevant core PHI technical competencies for this domain are:

Competency 3:

Considers **contextual factors** in the analysis of PHI data



Competency 4:

Contributes to a culture of **continuous quality improvement** of PHI data



Competency 8:

Demonstrates **effective use of data** for generating actionable insights



Courses

There is one separate course and one overarching course in this technical domain. The separate course has several learning objectives (Table 4), which collectively target the competencies that learners gain in this technical domain:

Overarching course T3.0 – communicate, discuss and document

Learners explore how to maintain communication with internal and external stakeholders to seek consultation on and discuss appropriate methods and the results of the analysis. Learners discover the importance of maintaining regular documentation of the results of the analysis.

Course T3.1 – conduct analysis to support risk assessment, decision-making and response

Learners explore how to interpret the data gathered considering the context (e.g. geopolitical incidents, cultural and sociological context) and conducting additional analyses, as appropriate. This may also require consulting technical experts in specific fields.



Table 4: Courses and learning objectives in technical domain - analyse

Courses	
Communicate, discuss and document	
Learning objectives	Describe how to present and discuss analytical results with internal and external stakeholders Explain the importance of continuous documentation of results
Conduct analysis to support risk assessment, decision-making, and response	
Learning objectives	Identify available and appropriate sources of information informing risk assessment, decision-making and response
	Identify epidemiological statistics that are appropriate for describing the nature and severity of different types of signals
	Describe the spatial analysis tools and techniques that can be used when analysing signals
	Apply epidemiological and statistical principles to interpret trends and insights from multiple information sources
	Apply epidemiological and statistical principles and techniques to identify populations and subgroups at increased risk
	Outline mathematical modelling principles and techniques to predict changes in the signal related to changes in other factors that encompass the all-hazards and One Health approaches
	Explain the limitations and uncertainties of epidemiological analyses, models, forecasts or other PHI data analysis methods
	Identify situations when additional expertise may be necessary to interpret analysis results
	Explain how different geopolitical events and sociological and cultural factors could impact PHI data and the results of analysis
	Interpret intelligence in the context of similar signals and the current context (e.g., geopolitical incidents, cultural and sociological context)
	Interpret new information about signals in the context of previous analyses

Technical Domain 4: Assess



Introduction



This topic focuses on equipping learners with the skills to assess the possible implications when a signal has been verified and classified as an event. Learners explore how to identify appropriate sources of information beyond surveillance data that are important for risk assessment. Learners discover how to critically assess analysis results to determine the significance of the event and the risk to affected people and populations, with reference to local contexts such as health-care capacity. Learners explore how to summarize the assessment results with an emphasis on close communication within and outside the organization. This process helps determine the risk level of the event, informing the next steps that should be taken.

Competencies

The most relevant core PHI technical competencies for this domain are:

Competency 7:

Takes a **solutions-oriented approach** to problem-solving in PHI



Competency 8:

Demonstrates **effective use of data** for generating actionable insights



Competency 12:

Learns from a range of **multisectoral One Health partners**



Competency 13:

Incorporates a **holistic approach** to PHI



Courses

There are two separate courses and one overarching course in this technical domain. Each course has several learning objectives (Table 5), which collectively target the competencies that learners gain in this technical domain:

Overarching course T4.0 – communicate, discuss and document

Learners focus on effective internal and external communication practices. Learners explore how to seek guidance and share information to refine the perceived importance of the event, adjust its scale, or integrate newly acquired data. Emphasis is placed on continuous documentation of new findings, including key insights gained through stakeholder communication. Learners develop skills to ensure all relevant updates are recorded accurately and used effectively to inform decision-making processes.

Course T4.1- conduct risk assessment for an event from PHI data

Learners delve into the intricacies of risk assessment. Learners identify and access appropriate sources of information crucial for comprehensive risk assessment beyond surveillance data, while distinguishing between evidence-based facts and assumptions. Learners discover how to engage One Health stakeholders to gather essential additional information for risk assessment. The topic also addresses the risk assessment tools and risk management options, considering interventions that alter the trajectory of the event while evaluating their potential impact on various sectors, including health systems, animal health, the environment, the economy and politics.

Course T4.2- generate insights for decision-making

Learners discover how to critically assess analysis results for generating insights and understanding the implications and potential outcomes of analyses. Special emphasis is placed on synthesizing analysis results into clear and coherent information tailored to diverse audiences. Learners explore techniques for presenting information logically, concisely and effectively to facilitate understanding and decision-making.

Table 5: Courses and learning objectives in technical domain 4 - assess

Courses	
Communicate, discuss and document	
Learning objectives	Apply the principles of effective communication and documentation to convey the results of a signal analysis
	Apply established documentation practices for the results of signal analysis and risk assessment
	Explain how to ensure compliance with reporting regulations in the context of assessing analysis results
Conduct risk assessment for an event from PHI data	
Learning objectives	Identify appropriate sources of information, beyond surveillance, needed for risk assessment
	Explain procedures for risk assessment of verified signals (events)
	Demonstrate how to formulate risk questions
	Explain how economic, environmental, political and social determinants of health affect populations
	Apply a risk assessment tool to assess the estimated level of risk
	Identify situations where a joint risk assessment tool may have to be applied
	Assess the relationship between the risk and the scale of action required
	Estimate the potential consequences of insufficient or untimely action
	Apply critical thinking to determine the level of risk and risk management options that are well reasoned, ethical, evidence-based, feasible and timely
	Describe how interventions should change the trajectory of the event
	Explain how interventions could affect other sectors, including health systems, animal health, the environment, the economy and politics
Generate insights for decision-making	
Learning objectives	Explain the features of a public health event that support decision-making and risk assessment
	Critically assess the results of the analysis to generate insights into the event
	Synthesize results into information that can be understood by diverse audiences
	Present information clearly, coherently, concisely and logically organized
	Prepare response recommendations for decision-makers that are commensurate with the results of the analysis and risk assessment

Functional Domain



Functional Domain: Collaborative Information Exchange



Introduction



This course focuses on effective strategies for collaboration and information exchange in PHI work, with an emphasis on functional competencies. Learners explore how to promptly communicate details about verified public health events to facilitate decision-making, alert stakeholders about unfolding events, and provide ongoing situational awareness and risk communication during emergencies. Learners discover how to ensure professional integrity, ethical standards, cultural sensitivity and adaptability in all collaborative efforts, enabling PHI professionals to navigate diverse situations and respond effectively to changing circumstances.

Competencies

All 16 core PHI functional competencies are applicable for this functional domain, but the following are the most relevant:

Competency 14:
Ensures others have
**clear, accurate, and
timely information**



Competency 15:
Pro-actively
**manages
interactions** with
others



Competency 18:
**Conveys
information**
purposefully



Competency 19:

Acknowledges the **importance of stakeholder information** needs

**Competency 20:**

Tailors communication to the audiences

**Competency 21:**

Works within the limits of **competence and scope of practice** for PHI activities

**Competency 25:**

Operates with **cultural sensitivity, respectfulness, and compassion**

**Competency 26:**

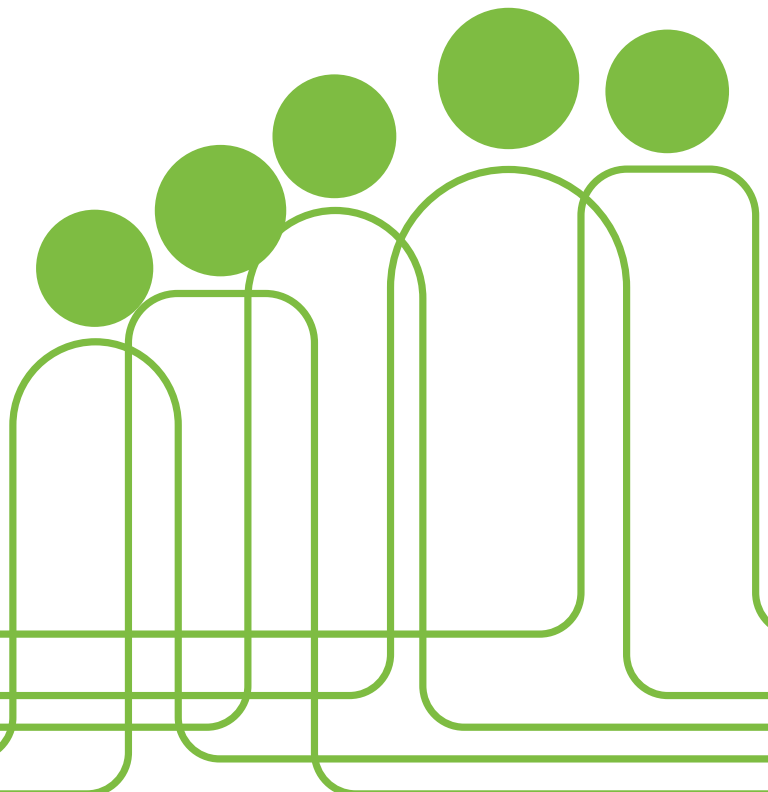
Adapts to **unexpected or changing situations**

**Competency 27:**

Plans strategically

**Competency 28:**

Empowers others



Courses

There are four courses in this functional domain. Each course has several learning objectives (Table 6), which collectively target the competencies that learners gain in this technical domain:

Course F1.1 – manage and enhance information exchange with stakeholders

This course focuses on honing skills that ensure effective information exchange to stakeholders across various contexts. Learners explore how to clarify goals and select appropriate information-sharing and communication tools to engage stakeholders. Training includes identification of barriers to information exchange, considering cultural context, technical knowledge and preferences.

Course F1.2 – provide timely intelligence of verified signals to support decision- making during a public health event

This course provides knowledge and skills for gathering information related to verified signals, ensuring information is consistent, and for tailoring risk communication specifically to meet the needs of various audiences, ensuring clarity, engagement and appropriate response actions. Learners discover how to identify and analyse target audiences, use modern communication tools and software, develop effective communication strategies, and assess the impact of their communications. Learners finish with a deeper understanding of crafting and disseminating messages during health emergencies and with practical tools and strategies to apply in their professional roles, ensuring their communications are effective, inclusive and responsive to the needs of their audiences.

Course F1.3 – adapt to change and challenges in PHI work

This course covers the importance of being adaptable in a dynamic field, allowing people to modify their approaches, strategies and practices in the context of evolving public health needs, new data, emerging trends and unexpected challenges. It emphasizes the importance of continuous professional development and self-care to enhance performance, maintain well-being and manage the demands of PHI work. Through these tasks, PHI professionals can ensure their work remains relevant, timely and effective in addressing public health issues, while preserving their own health.

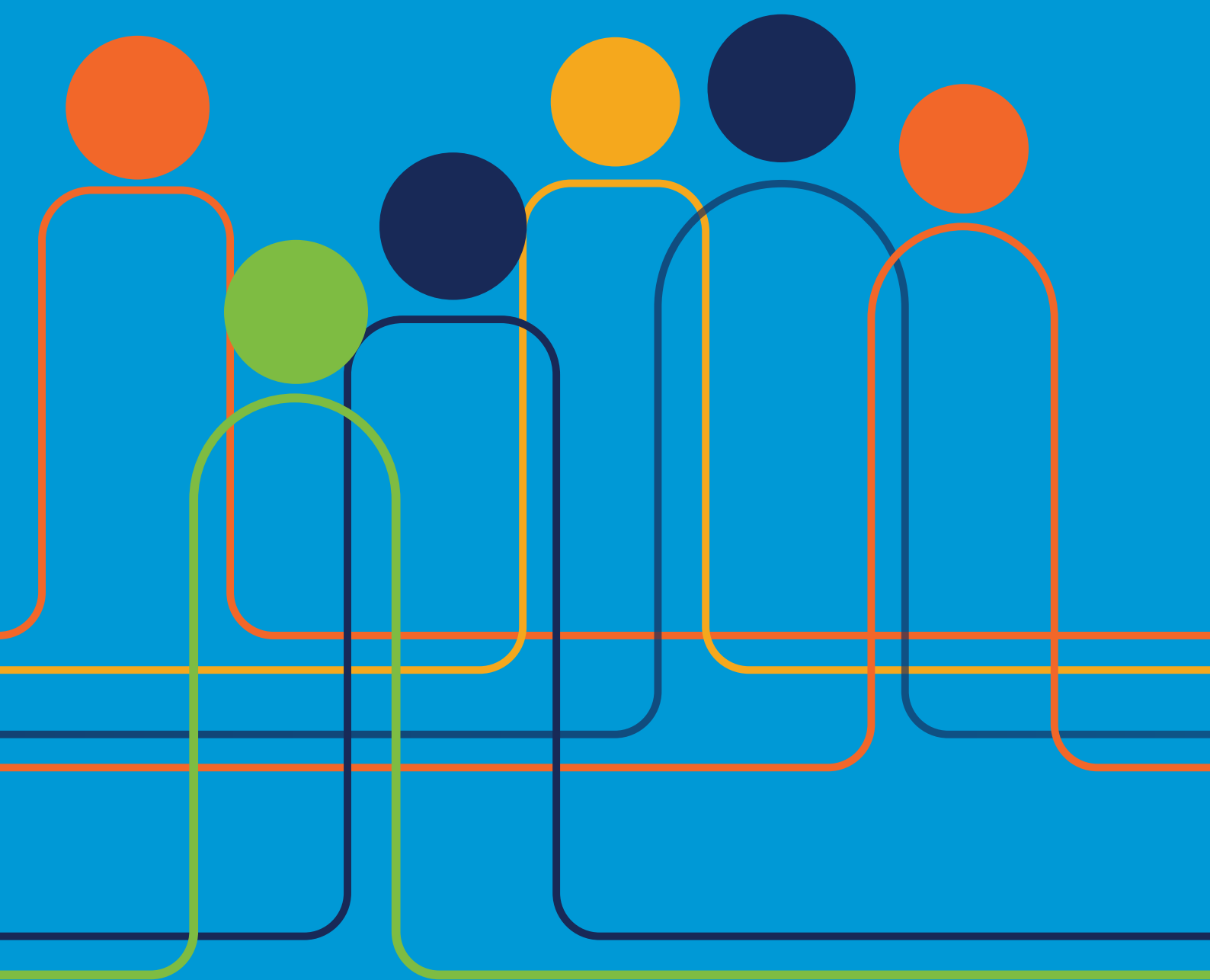
Course F1.4 – ensure successful collaboration and ethical practice in PHI work

This course covers effective PHI operations to ensure people work within their defined scope of competence, seek guidance when necessary, and adhere to ethical principles. It also focuses on fostering collaboration among stakeholders, with an emphasis on respecting cultural differences and diversity and empowering teams to perform at their best. Overall, this course helps to ensure PHI efforts are executed well, are responsive to evolving needs, and contribute to the achievement of measurable health improvements.

Table 6: Courses and learning objectives in functional domain – collaborative information exchange

Courses	
Manage and enhance information exchange with stakeholders	
Learning objectives	Define common goals and objectives for information exchange
	Describe barriers to effective information sharing
	Apply strategies that address barriers to effective information sharing
	Describe how to initiate communication with relevant stakeholders to seek feedback and provide critical information
	Explain how different types of PHI data can be integrated to deliver relevant, valuable and context-specific information
	Explain how different types of PHI data can be integrated to deliver relevant, valuable and context-specific information
	List ways to support others in effective information exchange
Provide timely intelligence of verified signals to support decision-making during a public health event	
Learning objectives	Identify which stakeholders need to be alerted for a given public health event
	Describe the critical intelligence needed to support decision-making
	Identify relevant ethical, privacy, confidentiality and legal requirements for obtaining, recording, storing, sharing, retaining and destroying information
	Explain how to evaluate stakeholder preferences, cultural contexts, awareness levels and technical knowledge
	Create a structured plan for sharing intelligence
	Describe which PHI highlights to include in a summary of the event
	Create a structured plan for responding to stakeholder requests for additional information
Adapt to change and challenges in PHI work	
Learning objectives	Identify information sources on emerging trends and technologies in PHI
	Identify relevant learning opportunities
	Describe approaches to identify change and challenges in PHI
	Evaluate new or unforeseen situations in PHI activities using critical thinking skills
	Apply problem-solving strategies during challenges or changes in PHI work
	Apply strategies to balance professional responsibilities with personal well-being to safeguard health and mental resilience
Ensure successful collaboration and ethical practice in PHI work	
Learning objectives	Apply PHI principles and the organization's mission to create a shared vision and key values for collaboration with stakeholders
	Identify realistic best practices for collaboration strategies that align with PHI
	Describe how to identify situations where tasks exceed current competence
	Describe approaches for seeking guidance from experts when tasks exceed current competence
	List strategies for overcoming barriers in PHI work due to disabilities or impairments
	Describe methods for addressing gender issues in PHI work
	Describe how to identify methods and strategies for promoting credibility and trust in health authorities



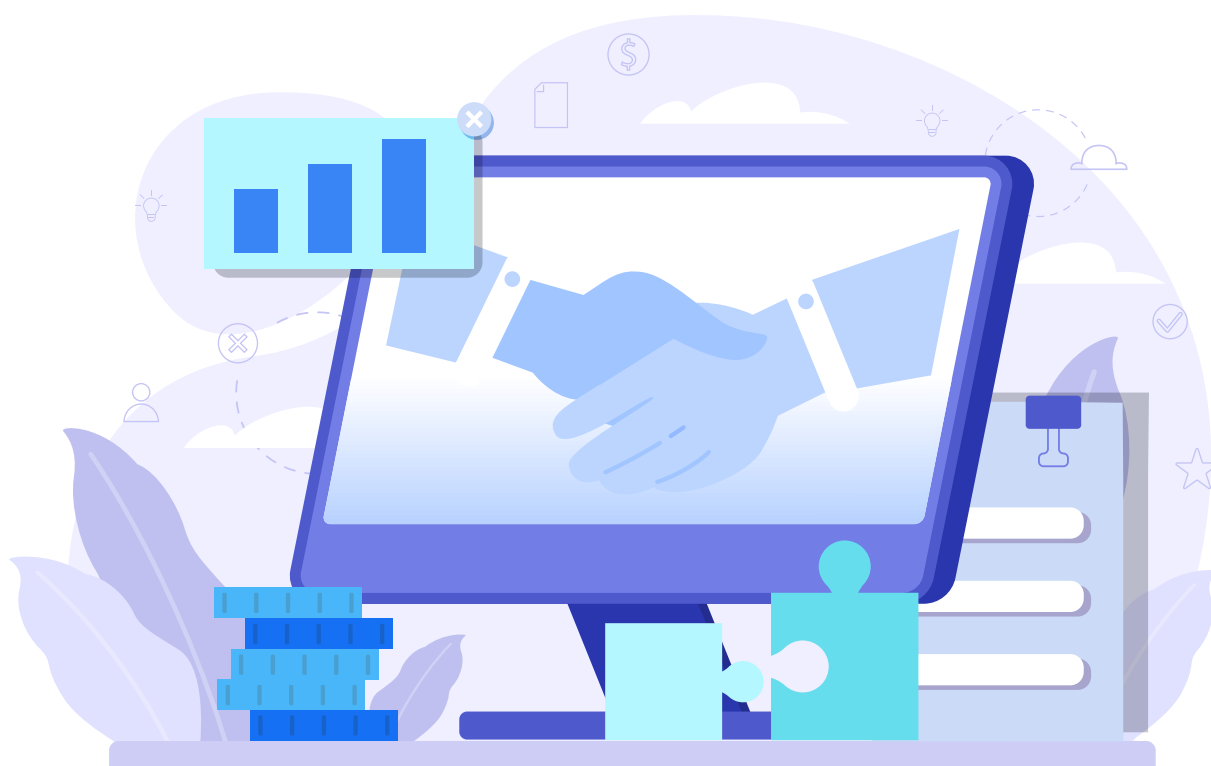


Concluding Remarks

The Public health intelligence curriculum represents a significant milestone in strengthening global health security by supporting the development of a competent and prepared PHI workforce. Aligned with the Public health intelligence competency framework, the Curriculum provides a structured, flexible, and practical tool to guide training efforts across diverse settings, roles, and levels of expertise. Through training that is grounded in a modular, all-hazards and multisectoral approach, including One Health principles, the Curriculum ensures relevance in a complex and evolving global health landscape.

This first-release version acknowledges the dynamic nature of PHI where methods, technologies, and threats continue to evolve. Therefore, the Curriculum is intended to be a living document, adaptable to future needs, contextual realities, and lessons learned through implementation. As countries and organizations adopt and tailor this Curriculum, it is expected to foster improved collaboration, coordination, and performance in PHI functions, ultimately enhancing preparedness, early detection, and timely response to health threats.

The successful application of this Curriculum will depend on sustained commitment to workforce development, institutional support for training initiatives, and ongoing refinement based on user feedback and emerging public health priorities. Together with the PHI Competency Framework and other global efforts, the PHI Curriculum offers a strong foundation for building resilient, informed, and action-oriented public health systems.



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Annex 1: Training in Foundations of Public Health Intelligence

Introduction



This comprehensive introductory course delves into the intricacies of public health intelligence (PHI), equipping learners with the foundational knowledge, skills and tools necessary to perform PHI activities. Learners explore the historical foundations, practical applications, ethical considerations and legal frameworks that underpin PHI activities. From understanding epidemiological features to engaging stakeholders diplomatically and navigating legal and ethical landscapes, the course covers essential topics to prepare learners for roles in PHI.

Competencies

This training has not been designed to cover specific competencies, but it contributes to building important parts of the following core technical and functional competencies:

Competency 3:

Considers **contextual factors** in the analysis of PHI data



Competency 4:

Contributes to a culture of **continuous quality improvement** of PHI data



Competency 7:

takes a **solutions-oriented approach** to problem-solving in PHI



Competency 9:

adapts priorities based on **unexpected or changing situations**



Competency 10:

Engages multisectoral **One Health stakeholders** for implementing joint PHI activities



Competency 11:

Builds and maintains a **trusting network** of multisectoral One Health stakeholders to support PHI activities



Competency 12:

Learns from a range of **multisectoral One Health partners**



Competency 13:

incorporates a **holistic approach** to PHI



Competency 15:

Proactively **manages interactions** with others



Competency 16:

Adapts communication to the goals, needs, cultural context, **urgency and sensitivity of the interaction**



Competency 17:

Listens **actively and attentively**



Competency 19:

Acknowledges the **importance of stakeholder information** needs



Competency 21:

Works within the limits of **competence and scope of practice** for PHI activities



Competency 22:

Engages in **lifelong learning and reflective practice** to improve performance of PHI activities



Competency 24:

Focuses on well-being, **recognizes stress and seeks support** when needed



Competency 26:

Adapts to **unexpected or changing situations**



Courses

There are eight courses in this training. Each course has several learning objectives (Table A1.1), which collectively target the competencies that learners gain by completing this training:

Course FP1.1 – evolution and background of PHI

This provides an overview of the history of PHI, the role of PHI in global health security over time, the basics of PHI, and the crucial role of PHI in global health security. Learners gain an understanding of the evolution of PHI, key PHI terminology, and the application of PHI in addressing diverse public health threats.

Course FP1.2 – all-hazards and One Health approaches to PHI

This provides an overview of the all-hazards and One Health approaches to PHI, highlighting the interconnectedness of humans, animals and the environment. It introduces the different types of public health hazards. Learners gain an understanding of all-hazards and One Health approaches and how they can be applied to PHI.

**Course FP1.3 –
legal, ethical
and regulatory
frameworks for PHI**

This provides an overview of the legal and ethical frameworks that govern PHI activities at the local, national and international levels. Learners gain an understanding of reporting obligations under local and international laws, including the International Health Regulations (2005), and explore the legal, regulatory, policy, governance and ethical considerations that shape PHI operations. The course addresses privacy, confidentiality, data protection requirements, and key principles of public health ethics that guide PHI practice.

**Course FP1.4 –
sources of
information and
the PHI process**

This provides a comprehensive exploration of information sources that can be used for PHI and describe the PHI process. Learners gain a broad understanding of information sources, including their value to PHI and common errors, limitations and biases that they may have. Learners gain insights into the steps of the PHI process and the activities conducted at each step.

**Course FP1.5 –
implementing
PHI**

This provides a comprehensive exploration of tools and methodologies that are essential for effective PHI activities. Learners discover how to modify PHI activities for different purposes and mitigate barriers in PHI activities. Learners explore different tools and resources that can be used for conducting PHI activities, including contemporary software and artificial intelligence.

**Course FP1.6 –
epidemiology
for PHI**

This covers public health threats, their epidemiological characteristics, and the factors that contribute to their complexity and dynamic nature. Learners explore how economic, environmental, political and social determinants affect the impact of public health threats, gaining insight into how these factors shape health outcomes. Learners discover different surveillance approaches, including collaborative and integrated surveillance, to enhance timely detection of public health threats.

Course FP1.7 – PHI during public health emergencies

This offers an overview of public health emergencies and the role of PHI in preparedness, response and recovery efforts. Learners gain an understanding of the common types of public health emergencies, and the impact they have on populations and the public health sector. Learners are introduced to self-care practices and building resilience in the context of health emergencies.

Course FP1.8 – stakeholder engagement and communication

This focuses on developing effective communication skills and stakeholder engagement strategies essential for PHI activities. Learners explore methods to identify and engage key stakeholders, emphasizing the importance of collaboration in PHI. The course covers best practices for diplomatic and effective communication with stakeholders, team members and leadership, ensuring respectful and productive interactions.

Table A1. Courses and learning objectives in foundations of public health intelligence

Courses	
Evolution and background of PHI	
Learning objectives	Summarize the history of PHI and epidemic intelligence
	Define PHI and key PHI terminology
	Describe health security and the role PHI plays in health security
	Describe PHI in the context of the International Health Regulations (IHR 2005)
All-hazards and One Health approaches to PHI	
Learning objectives	Distinguish common public health hazards
	Explain the all-hazards approach to PHI
	Explain the One Health approach to PHI
	Describe the interconnectedness of human, animal and environmental health and how changes in one could affect the others
Legal, ethical and regulatory frameworks for PHI	
Learning objectives	Identify local and international laws, including IHR, that have reporting obligations
	Outline national, regional and international public health legislation, regulations, policy, governance and ethical frameworks for operating PHI
	Outline how the IHR operate at the local, national and international levels
	Identify relevant ethical, privacy, confidentiality and legal requirements for obtaining, recording, storing, sharing and destroying acquired information
	Describe what information is confidential
	Describe public health ethics

Courses	
Sources of PHI information and the PHI process	
Learning objectives	List types of information sources that can be used for PHI activities
	Explain indicator-based surveillance and event-based surveillance data sources in terms of their value for PHI
	Identify common errors, limitations and biases of information sources used for PHI activities
	Explain approaches for receiving signals from a variety of information sources
	List the steps of the PHI process
	Describe the purpose of and the activities conducted during each step of the PHI process
	Describe how PHI steps relate to each other
	Explain the importance of documenting signals, actions taken and outcomes
	Choose which PHI step(s) need to be conducted based on available information and context
Implementing PHI	
Learning objectives	Explain how PHI activities should be modified for different types of public health hazards
	Describe how to integrate the mission and priorities of the public health organization into PHI activities
	Describe ways to mitigate the causes and consequences of discrimination, exclusion, prejudice, stigma and other barriers to healthy communities in PHI activities
	List tools and resources that can be used to perform PHI activities
	List examples of contemporary software and systems for managing, analysing and reporting data and information
	Explain the use of artificial intelligence (AI) tools for PHI activities
Epidemiology for PHI	
Learning objectives	Identify the epidemiological features of health threats using diverse sources of information
	Describe the complexity and dynamic nature of health threats
	Explain how economic, environmental, political and social determinants of health affect public health threats
	Explain the relationship between surveillance and PHI
	Describe collaborative surveillance and integrated surveillance
PHI during public health emergencies	
Learning objectives	Define public health emergencies
	List common types of public health emergencies
	Describe the impact of public health emergencies on populations and the public health sector
	Explain the role and functioning of PHI in public health emergencies
	Explain how routine surveillance and other sources of information may be affected during an emergency
	List self-care strategies that promote emotional resilience, health and well-being
	List strategies to manage fatigue, ill health, stress and the impact of exposure to distressing and emergency situations
Stakeholder engagement and communication	
Learning objectives	Explain how to engage stakeholders who can contribute to PHI activities
	Explain the importance of collaborative decision-making
	Demonstrate diplomacy in communications with stakeholders, team and leadership
	Analyse the other parties involved in interactions
	Identify the appropriate style, language and method of communication for the interaction
	Demonstrate empathy, respect and tact during communication

Annex 2: Technical PHI Competencies



Competency 1: evaluates data and information from a range of sources

- 1.1 Critically appraises the limitations, quality, relevance and significance of data, information and evidence for PHI
- 1.2 Demonstrates critical thinking to synthesize information from multiple sources
- 1.3 Demonstrates initiative to prevent the inclusion of misinformation in PHI activities
- 1.4 Promotes access to data, information and evidence for PHI within and across sectors, organizations, fields and disciplines
- 1.5 Recognizes when there is a need for additional data and information
- 1.6 Considers special characteristics of the technology that generated the data and information
- 1.7 Seeks data, information and evidence from a range of multisectoral sources for PHI activities



Competency 2: engages in practices to ensure data security, confidentiality, and privacy

- 2.1 Acts ethically in the collection, use and analysis of PHI data
- 2.2 Upholds policies for maintaining the integrity and security of data used for PHI activities
- 2.3 Upholds privacy, confidentiality and legal requirements for obtaining, recording, storing, sharing, retaining and destroying acquired information



Competency 3: considers contextual factors in the analysis of PHI data

- 3.1 Challenges own biases in the analysis of PHI data
- 3.2 Considers geopolitical events and trends on PHI data
- 3.3 Considers the sociological, environmental and cultural context in which PHI data are collected
- 3.4 Considers the temporal context in which PHI data are collected



Competency 4: contributes to a culture of continuous quality improvement of PHI data

- 4.1 Adheres to data verification requirements for PHI data
- 4.2 Contributes to quality measurement and continuous quality improvement processes for PHI data
- 4.3 Learns from what works and what has not gone well in collection and analysis of PHI data
- 4.4 Identifies types of corroborating evidence needed to improve confidence or make decisions
- 4.5 Offers suggestions for improvement to address identified problems with PHI data
- 4.6 Promotes and implements quality assurance techniques for PHI data



Competency 5: takes an adaptive, collaborative and rigorous approach to PHI decision-making

- 5.1 Adapts the approach to decision-making that reflects the complexity and urgency of the situation
- 5.2 Approaches decisions analytically and methodically
- 5.3 Considers the consequences of decisions
- 5.4 Demonstrates critical thinking to reach decisions that are well reasoned, ethical, evidence-informed, feasible, timely and based on the best available PHI information
- 5.5 Evaluates assumptions against facts
- 5.6 Promotes collaborative decision-making
- 5.7 Seeks information and evidence from a range of multisectoral sources when approaching decision-making



Competency 6: incorporates a systems approach to decision-making, considering One Health and all-hazards perspectives

- 6.1 Acknowledges the complex nature of health threats
- 6.2 Anticipates the consequences of decisions on other sectors
- 6.3 Conceptualizes interrelationships between elements considering One Health and all-hazards perspectives



Competency 7: takes a solutions-oriented approach to problem-solving in PHI

- 7.1 Creates pragmatic solutions to identified problems in PHI activities
- 7.2 Focuses on solutions, end goals and results
- 7.3 Takes initiative to mitigate anticipated problems in PHI activities



Competency 8: demonstrates effective use of data for generating actionable insights

- 8.1 Ensures the use of timely and relevant PHI data to inform action
- 8.2 Seeks feedback on the value of data, insights and reports
- 8.3 Promotes the use of technology to generate insights



Competency 9: adapts priorities based on unexpected or changing situations

- 9.1 Adjusts priorities to respond according to the evolution of the public health events and demands
- 9.2 Embraces the dynamic nature of health threats
- 9.3 Manages competing demands



Competency 10: engages multisectoral One Health stakeholders for implementing joint PHI activities

- 10.1 Celebrates shared outcomes, goals and values across multiple sectors, fields and geographies when relevant
- 10.2 Enables multisectoral stakeholders to make contributions to PHI activities
- 10.3 Engages with stakeholders across sectors, organizations, fields and disciplines who can contribute to or influence the effectiveness of PHI activities
- 10.4 Adheres to the established and agreed ways of working within PHI activities
- 10.5 Jointly negotiates roles and responsibilities for effective and efficient PHI activities



Competency 11: builds and maintains a trusting network of multisectoral One Health stakeholders to support PHI activities

- 11.1 Maintains constructive and collaborative working relationships with stakeholders through both formal and informal channels
- 11.2 Maintains ethical boundaries with other members of the PHI network
- 11.3 Minimizes the impact hierarchical differences between stakeholders, including different levels of governance and sectors
- 11.4 Strives to develop a positive rapport with stakeholders characterized by respect, support and trust



Competency 12: learns from a range of multisectoral One Health partners

- 12.1 Demonstrates willingness to learn from the experience of others
- 12.2 Engages in opportunities to improve collaboration within PHI teams and between multisectoral One Health partners
- 12.3 Learns from interactions with others and the feedback process
- 12.4 Provides constructive, sensitive and timely feedback and support to improve PHI activities
- 12.5 Seeks constructive, sensitive and timely feedback and support to improve PHI activities



Competency 13: incorporates a holistic approach to PHI

- 13.1 Considers behavioural and social science
- 13.2 Considers economic, environmental, political and social determinants of health
- 13.3 Contributes to protecting vulnerable populations
- 13.4 Recognizes the interconnectedness of humans, animals and the environment

Annex 3: Functional PHI Competencies



Competency 14: ensures others have clear, accurate and timely information

- 14.1 Shares information with relevant others in a timely manner
- 14.2 Supports others to speak about the progress of PHI activities and public health events
- 14.3 Promotes credibility and trust in health institutions



Competency 15: proactively manages interactions with others

- 15.1 Clarifies the communication goals for an interaction
- 15.2 Identifies when and how to initiate, conduct and close an interaction
- 15.3 Manages communication barriers due to cognitive, physical or sensory impairment, culture, developmental stage, geography or language
- 15.4 Supports others to communicate for themselves



Competency 16: adapts communication to the goals, needs, cultural context, urgency and sensitivity of the interaction

- 16.1 Adapts the style, language and method of communication to the interaction
- 16.2 Maintains an approach to communication characterized by calmness, compassion, empathy, respect, sensitivity and tact
- 16.3 Seeks to address gender issues in communication settings
- 16.4 Seeks to mitigate the impact of own beliefs, biases, emotional responses, opinions and values on verbal and non-verbal communication
- 16.5 Uses a range of verbal, non-verbal, visual, written and digital communication tools and techniques
- 16.6 Uses relevant abbreviations, language and terminology, translating complex and technical content into lay terms when necessary



Competency 17: listens actively and attentively

- 17.1 Responds sensitively to what others express
- 17.2 Supports others to ask questions and openly express experiences, feelings, ideas and opinions
- 17.3 Uses strategies to clarify what others express
- 17.4 Uses a range of non-verbal cues and verbal affirmations



Competency 18: conveys information purposefully

- 18.1 Adopts strategies that encourage a common understanding of information and decisions
- 18.2 Conveys quantitative and qualitative information of value and relevance to the audience
- 18.3 Differentiates between information as facts, context-specific evidence, opinion and misinformation
- 18.4 Expresses own opinions and perspectives with clarity, confidence and respect
- 18.5 Presents information clearly, coherently, concisely and organized logically
- 18.6 Provides information that supports the goals of the communication
- 18.7 Provides relevant, accurate and complete information



Competency 19: acknowledges the importance of stakeholder information needs

- 19.1 Communicates openly and honestly about what is known and not known
- 19.2 Creates opportunities for discussion
- 19.3 Demonstrates initiative to communicate with stakeholders



Competency 20: tailors communication to the audience

- 20.1 Accommodates audiences with diverse abilities to communicate
- 20.2 Develops and tailors messages for different audiences, using appropriate communication strategies, communication media and channels
- 20.3 Expresses information in a way that is suitable for the audience
- 20.4 Recognizes the capacity of the audience to comprehend technical information
- 20.5 Speaks and writes clearly and plainly
- 20.6 Strives to meet the audience's information needs



Competency 21: works within the limits of competence and scope of practice for PHI activities

- 21.1 Adheres to the duties, obligations and codes of conduct defined by occupational standards, legal regulations and organizational procedures for PHI activities
- 21.2 Maintains awareness of own competence and scope of practice for PHI activities
- 21.3 Seeks guidance when encountering situations beyond competence or scope of practice



Competency 22: engages in lifelong learning and reflective practice to improve performance of PHI activities

- 22.1 Engages in self-learning and reflective practice on PHI practice
- 22.2 Modifies own attitudes, behaviours and gaps in competence based on reflective practice
- 22.3 Seeks and engages in continuous formal and informal learning linked to current and emerging PHI practices



Competency 23: demonstrates high standards of ethical conduct

- 23.1 Acts with honesty, integrity and transparency
- 23.2 Avoids overuse or misuse of resources
- 23.3 Consults with others in situations with ethical implications
- 23.4 Honours commitments
- 23.5 Refuses individual gifts or other forms of influence intended to coerce or invite personal favour
- 23.6 Takes ownership of responsibilities and decisions, and their consequences
- 23.7 Uses physical, human and financial resources efficiently
- 23.8 Upholds legal and ethical principles for public health practice



Competency 24: focuses on well-being, recognizes stress and seeks support when needed

- 24.1 Engages in self-care practices that promote emotional resilience, health and well-being
- 24.2 Monitors mental, physical and social well-being in self and others
- 24.3 Understands how to identify signs of stress and ill health in self and others
- 24.4 Identifies strategies and resources available for support and help when needed for self and others
- 24.5 Seeks help or support where needed for own health and well-being
- 24.6 Uses a range of strategies to manage fatigue, ill health, stress and the impact of exposure to distressing and emergency situations



Competency 25: operates with cultural sensitivity, respectfulness, and compassion

- 25.1 Adopts an approach to practice that is non-blaming, non-discriminatory, non-judgemental and non-stigmatizing
- 25.2 Critically considers the causes and consequences of discrimination, exclusion, prejudice, stigma and other social challenges that influence trajectory of the threat
- 25.3 Demonstrates cultural sensitivity and embraces cultural diversity
- 25.4 Demonstrates compassion, empathy and respect for all groups of people, irrespective of age, asylum or migration status, criminal record, culture, disability, economic status, ethnicity, gender identity and expression, health literacy, health status, language, nationality, race, religion, sex, sexual orientation, treatment adherence, vulnerability to ill health, or any other characteristic
- 25.5 Maintains self-awareness around own beliefs, biases, emotional responses and values



Competency 26: adapts to unexpected or changing situations

- 26.1 Demonstrates a calm demeanour under pressure
- 26.2 Demonstrates flexibility to meet the needs of the situation
- 26.3 Uses a range of strategies to maintain PHI activities under pressure



Competency 27: plans strategically

- 27.1 Contributes to developing key values and a shared vision in planning and implementing policy
- 27.2 Contributes to maintaining organizational performance standards
- 27.3 Contributes to team and organizational learning to advance public health goals
- 27.4 Embraces the mission and priorities of the organization
- 27.5 Generates and communicates broad and compelling organizational direction, inspiring others to pursue the same direction
- 27.6 Proactively develops strategies to accomplish objectives
- 27.7 Uses public health ethics to manage self, others, information and resources



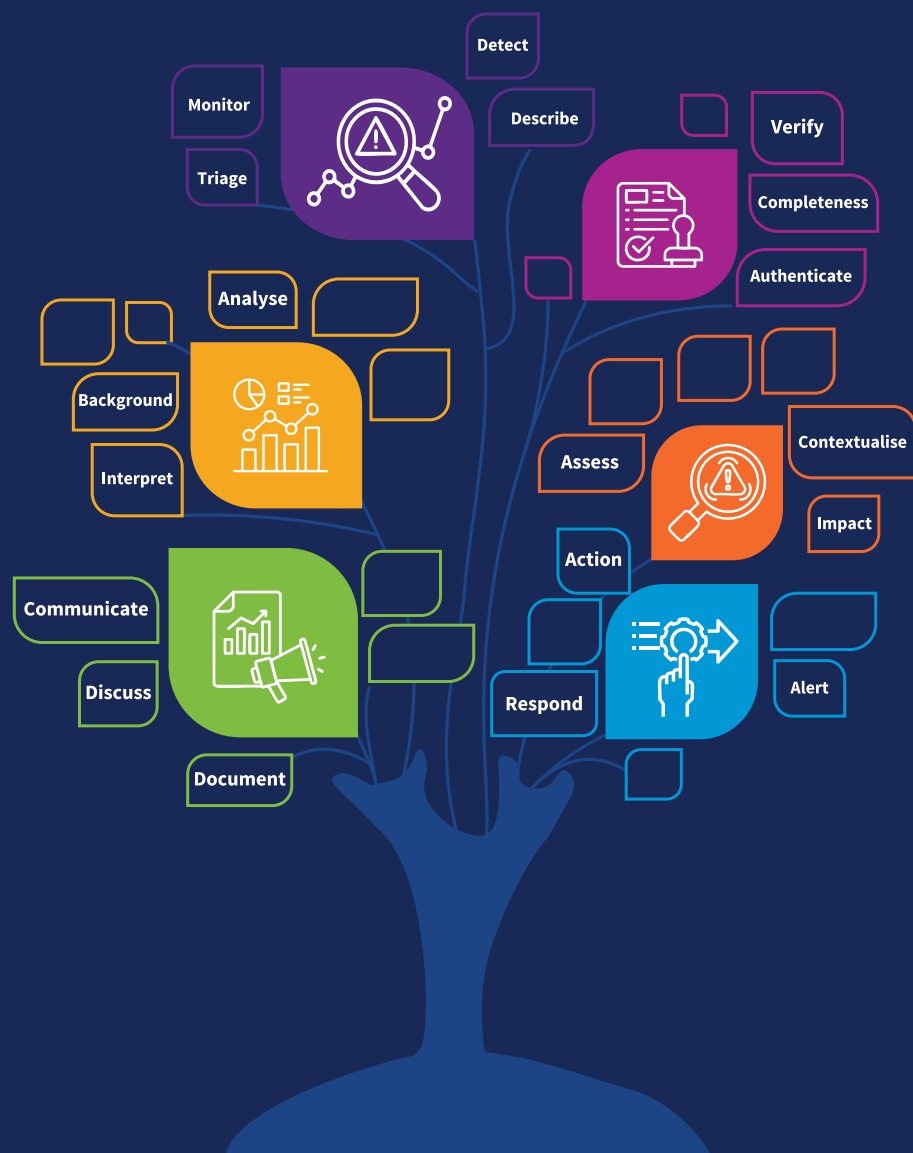
Competency 28: empowers others

- 28.1 Encourages contribution of others in translating vision into results
- 28.2 Encourages others to set challenging goals
- 28.3 Gives staff autonomy in important fields of their work
- 28.4 Holds others accountable for achieving results related to their field of responsibility
- 28.5 Shows appreciation for achievement and effort



Competency 29: manages tensions and conflicts

- 29.1 Anticipates tensions or potential areas of conflict
- 29.2 Considers different perspectives when seeking compromise, consensus or a decision
- 29.3 Focuses on the sources of tensions rather than arising conflicts
- 29.4 Supports a blame-free environment in which one is safe to question and seek support and guidance
- 29.5 Supports and acts in accordance with final group decision, even when such decisions may not entirely reflect own position
- 29.6 Takes positive action to avoid and dispel abuse, harassment or other disruptive behaviours
- 29.7 Uses diplomacy to mediate, negotiate or persuade
- 29.8 Serves as a role model



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